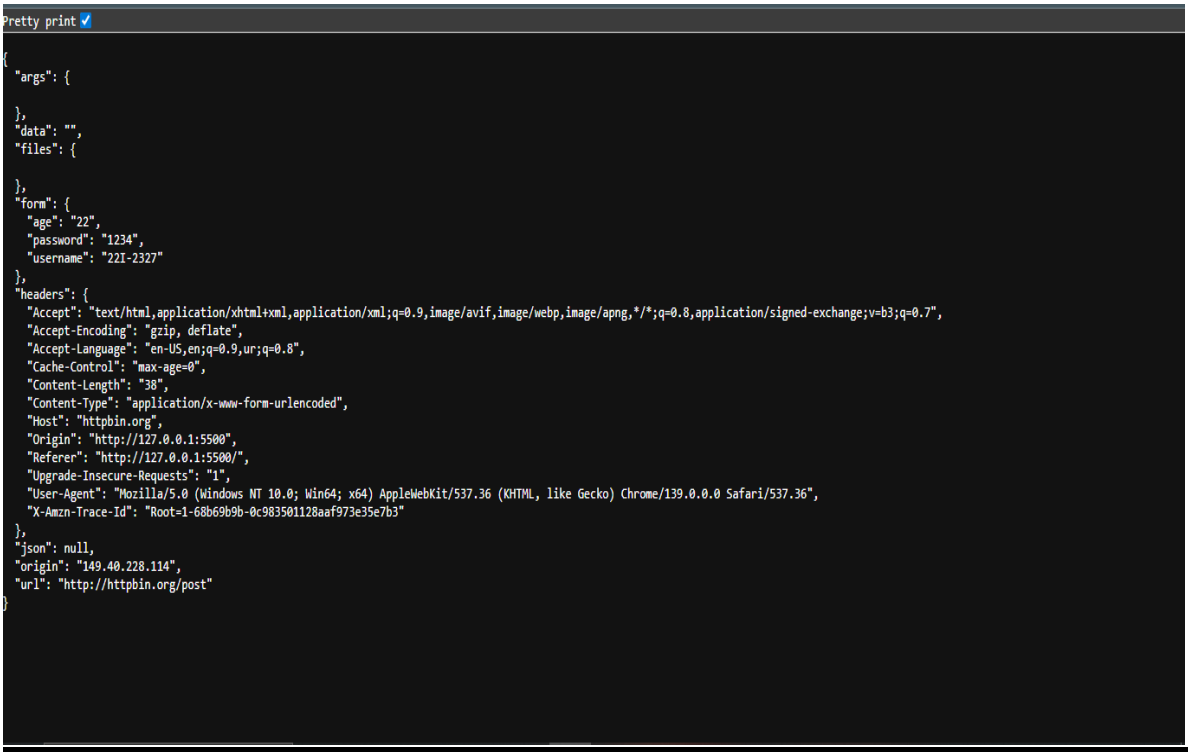
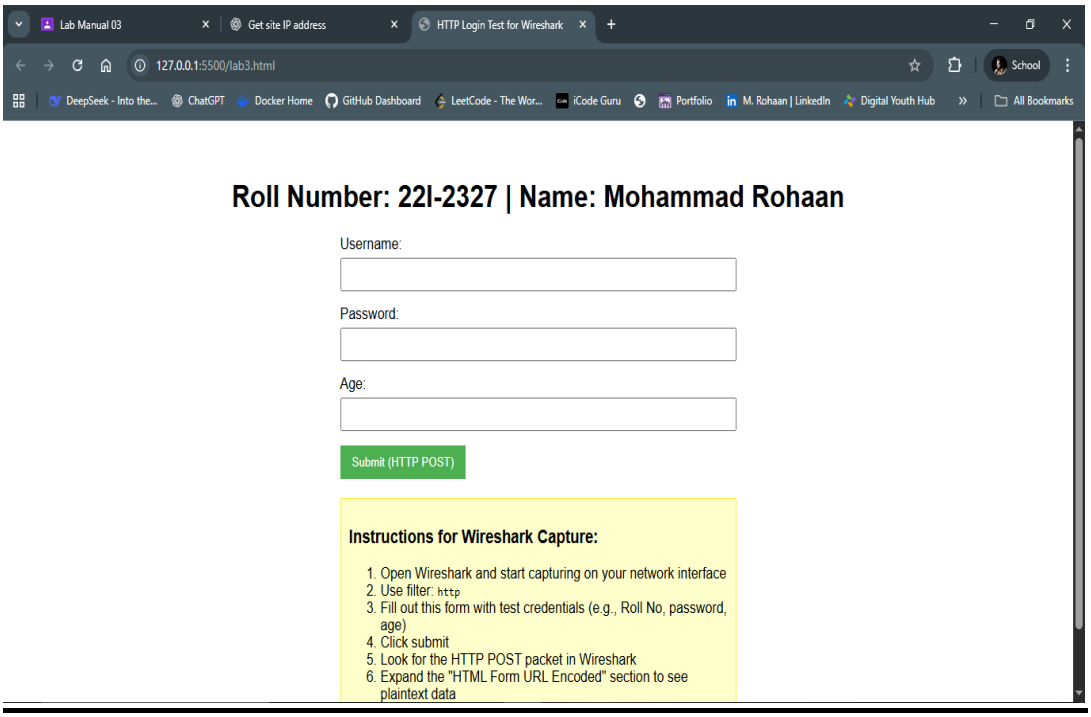
# Title: *“Computer Networks Lab – Wireshark Analysis”*

**Mohammad Rohaan 22I-2327 Section E Lab# 03**

## Task# 01:







# Roll Number: 22I-2327 | Name: Mohammad Rohaan

Username:

22I-2327

Password:

••••

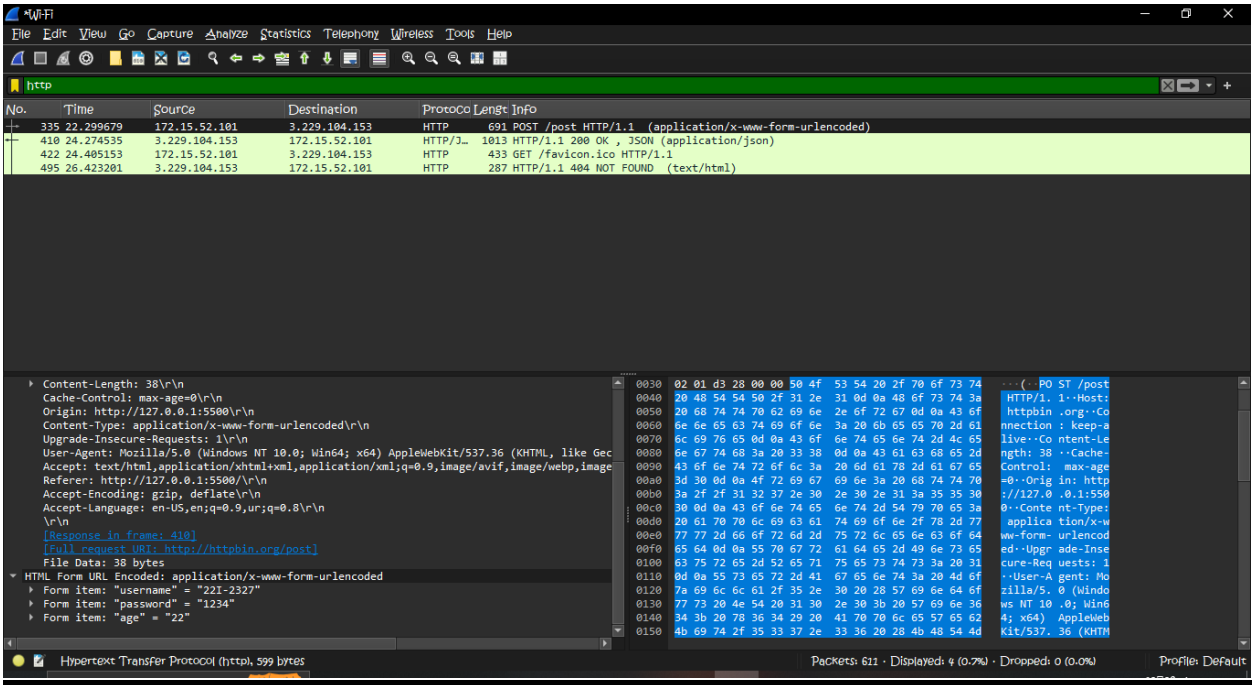
Age:

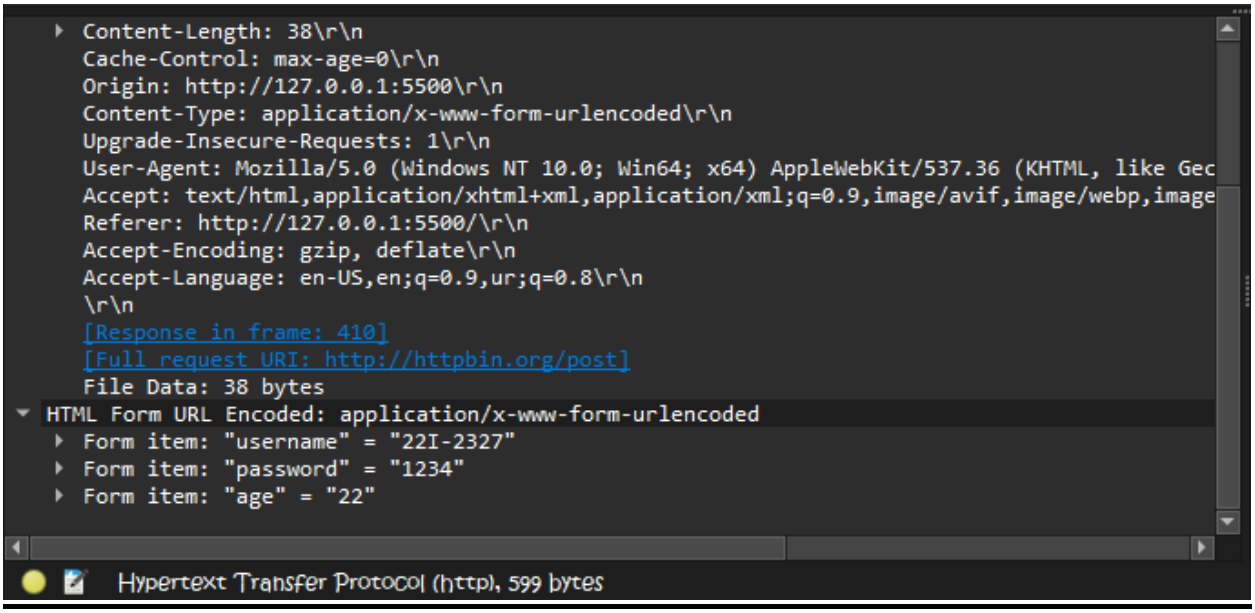
22

Submit (HTTP POST)

## Instructions for Wireshark Capture:

1. Open Wireshark and start capturing on your network interface
2. Use filter: `http`
3. Fill out this form with test credentials (e.g., Roll No, password, age)
4. Click submit
5. Look for the HTTP POST packet in Wireshark
6. Expand the "HTML Form URL Encoded" section to see plaintext data


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Source Destination Protocol Length Info
335 22.299679 172.15.52.101 3.229.104.153 HTTP 691 POST /post HTTP/1.1 (application/x-www-form-urlencoded)
410 24.274535 3.229.104.153 172.15.52.101 HTTP/J… 1013 HTTP/1.1 200 OK , JSON (application/json)
422 24.405153 172.15.52.101 3.229.104.153 HTTP 433 GET /favicon.ico HTTP/1.1
495 26.423201 3.229.104.153 172.15.52.101 HTTP 287 HTTP/1.1 404 NOT FOUND (text/html)
Content-Length: 38\r\n
Cache-Control: max-age=0\r\n
Origin: http://127.0.0.1:5500\r\n
Content-Type: application/x-www-form-urlencoded\r\n
Upgrade-Insecure-Requests: 1\r\n
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gec
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image
Referer: http://127.0.0.1:5500/\r\n
Accept-Encoding: gzip, deflate\r\n
Accept-Language: en-US,en;q=0.9,ur;q=0.8\r\n
\r\n
[Response in frame: 410]
[Full request URI: http://httpbin.org/post]
File Data: 38 bytes
HTML Form URL Encoded: application/x-www-form-urlencoded
Form item: "username" = "22I-2327"
Form item: "password" = "1234"
Form item: "age" = "22"
0030 02 01 d3 28 00 00 50 4f 53 54 20 2f 70 6f 73 74
0040 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a
0050 20 68 74 74 70 62 69 6e 2e 6f 72 67 0d 0a 43 6f
0060 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 70 2d 61
0070 6c 69 76 65 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 65
0080 6e 67 74 68 3a 20 33 38 0d 0a 43 61 63 68 65 2d
0090 43 6f 6e 74 72 6f 6c 3a 20 6d 61 78 2d 61 67 65
00a0 3d 30 0d 0a 4f 72 69 67 69 6e 3a 20 68 74 74 70
00b0 3a 2f 2f 31 32 37 2e 30 2e 30 2e 31 3a 35 35 30
00c0 30 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a
00d0 20 61 70 70 6c 69 63 61 74 69 6f 6e 2f 78 2d 77
00e0 77 77 2d 66 6f 72 6d 2d 75 72 6c 65 6e 63 6f 64
00f0 65 64 0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65
0100 63 75 72 65 2d 52 65 71 75 65 73 74 73 3a 20 31
0110 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f
0120 7a 69 6c 6c 61 2f 35 2e 30 20 28 57 69 6e 64 6f
0130 77 73 20 4e 54 20 31 30 2e 30 3b 20 57 69 6e 36
0140 34 3b 20 78 36 34 29 20 41 70 70 6c 65 57 65 62
0150 4b 69 74 2f 35 33 37 2e 33 36 20 28 4b 48 54 4d
···(··PO ST /post
HTTP/1. 1··Host:
httpbin .org··Co
nnection : keep-a
live··Co ntent-Le
ngth: 38 ··Cache-
Control: max-age
=0··Orig in: http
://127.0 .0.1:550
0··Conte nt-Type:
applica tion/x-w
ww-form- urlencod
ed··Upgr ade-Inse
cure-Req uests: 1
··User-A gent: Mo
zilla/5. 0 (Windo
ws NT 10 .0; Win6
4; x64) AppleWeb
Kit/537. 36 (KHTM
Hypertext Transfer Protocol (http), 599 bytes
Packets: 611 · Displayed: 4 (0.7%) · Dropped: 0 (0.0%)
Profile: Default


Content-Length: 38\r\n
Cache-Control: max-age=0\r\n
Origin: http://127.0.0.1:5500\r\n
Content-Type: application/x-www-form-urlencoded\r\n
Upgrade-Insecure-Requests: 1\r\n
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gec
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image
Referer: http://127.0.0.1:5500/\r\n
Accept-Encoding: gzip, deflate\r\n
Accept-Language: en-US,en;q=0.9,ur;q=0.8\r\n
\r\n
[Response in frame: 410]
[Full request URI: http://httpbin.org/post]
File Data: 38 bytes
HTML Form URL Encoded: application/x-www-form-urlencoded
Form item: "username" = "22I-2327"
Form item: "password" = "1234"
Form item: "age" = "22"
Hypertext Transfer Protocol (http), 599 bytes

Q1: Why is HTTP considered insecure for transmitting sensitive data?

Answer:
HTTP is unsafe because it sends data as plain text. This means whatever you type in a form (like username, password, or age) travels across the internet without any protection.

- Anyone can listen: Hackers on the same network (like public Wi-Fi) can use tools such as Wireshark to see exactly what you typed.
- Man-in-the-Middle (MITM): Attackers can intercept your messages and even change them before they reach the website.
- Password theft: Since your login details are visible, attackers can easily steal your account.
- Session hijacking: Hackers might grab your session info and pretend to be you.

How attackers exploit this:

They run a “packet sniffer” (like Wireshark or tcpdump) on the network. Because HTTP does not hide anything, they can directly read things like:

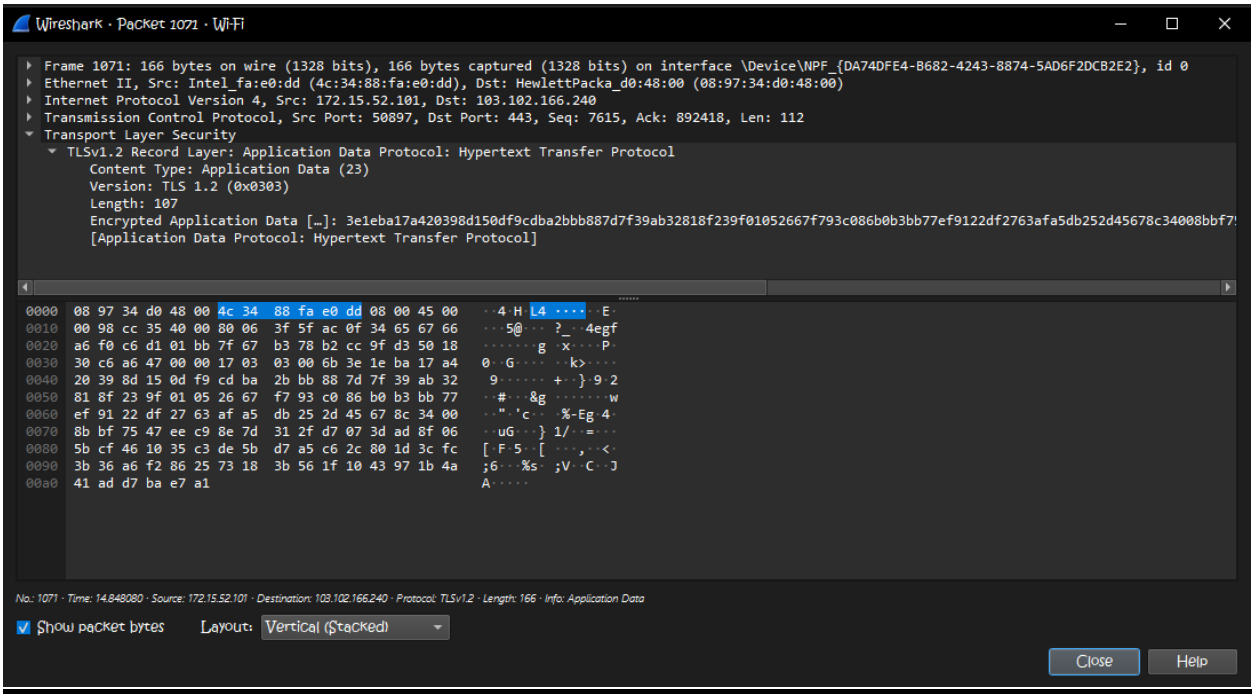
username=2327&password=1234&age=22

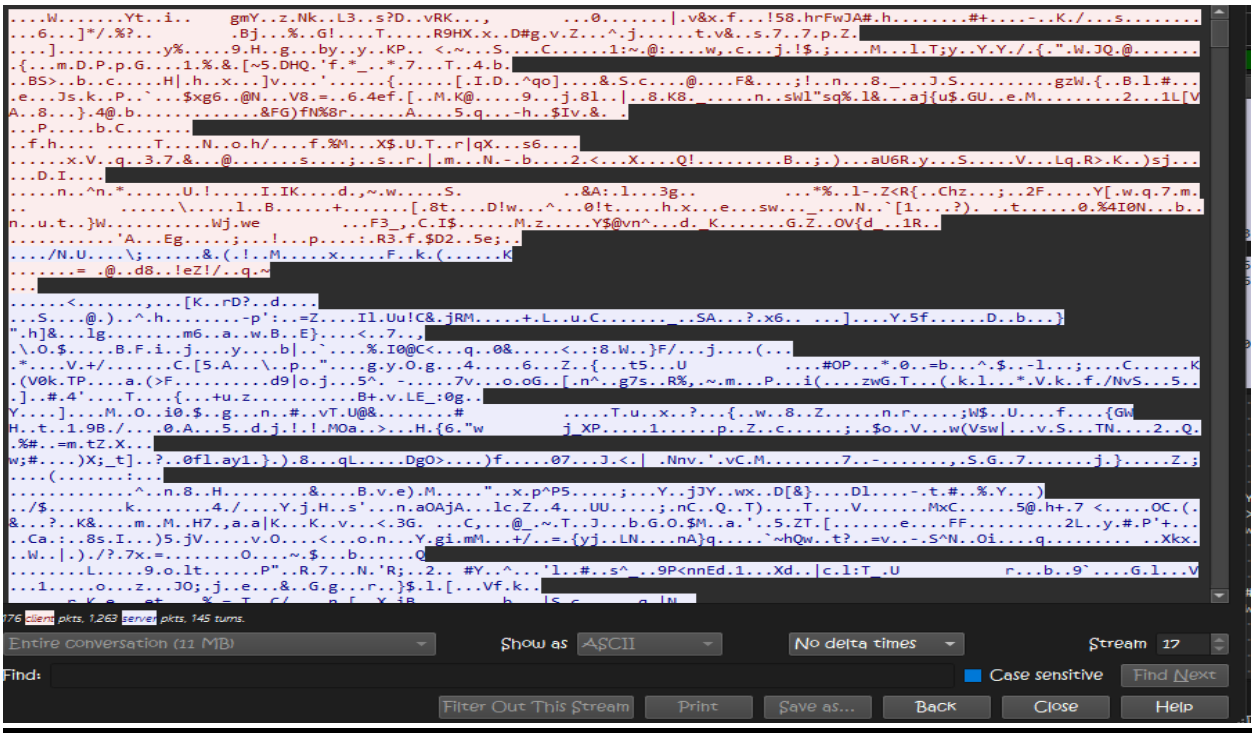
The Solution: HTTPS

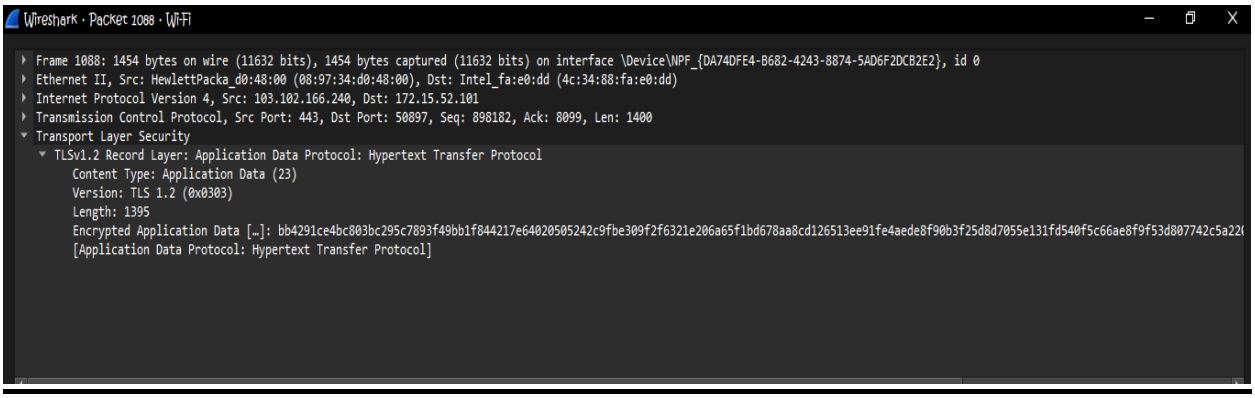
That’s why modern websites use HTTPS. The “S” means Secure. It uses SSL/TLS encryption, which scrambles the data so even if hackers capture it, they can’t understand it.

# Task# 02:


Wikipedia
The Free Encyclopedia
Search Wikipedia
Search
Donate
Create account
Log in
Wiki Loves Monuments: Photograph a monument, help Wikipedia and win!
Learn more
Tomato
195 languages
Contents
hide
(Top)
Naming
History
Botany
Plant breeding
Cultivation
Pests and diseases
As food
In popular culture
See also
References
Article
Talk
Read
View source
View history
Tools
Appearance
hide
Text
Small
Standard
Large
Width
Standard
Wide
Color (beta)
Automatic
From Wikipedia, the free encyclopedia
For other uses, see Tomato (disambiguation).
The tomato (US: /təˈmeɪtoʊ/, UK: /təˈmɑːtoʊ/; Solanum lycopersicum) is a plant whose fruit is an edible berry that is eaten as a vegetable. The tomato is a member of the nightshade family that includes tobacco, potato, and chili peppers. It originated from western South America, and may have been domesticated there or in Mexico (Central America). It was introduced to the Old World by the Spanish in the Columbian exchange in the 16th century.
Tomato plants are vines, largely annual and vulnerable to frost, though sometimes living longer in greenhouses. The flowers are able to self-
Tomato


Wireshark · Packet 1071 · Wi-Fi
Frame 1071: 166 bytes on wire (1328 bits), 166 bytes captured (1328 bits) on interface \Device\NPF_{DA74DFE4-B682-4243-8874-5AD6F2DCB2E2}, id 0
Ethernet II, Src: Intel_fa:e0:dd (4c:34:88:fa:e0:dd), Dst: HewlettPacka_d0:48:00 (08:97:34:d0:48:00)
Internet Protocol Version 4, Src: 172.15.52.101, Dst: 103.102.166.240
Transmission Control Protocol, Src Port: 50897, Dst Port: 443, Seq: 7615, Ack: 892418, Len: 112
Transport Layer Security
TLSv1.2 Record Layer: Application Data Protocol: Hypertext Transfer Protocol
Content Type: Application Data (23)
Version: TLS 1.2 (0x0303)
Length: 107
Encrypted Application Data […]: 3e1eba17a420398d150df9cdba2bbb887d7f39ab32818f239f01052667f793c086b0b3bb77ef9122df2763afa5db252d45678c34008bbf7
[Application Data Protocol: Hypertext Transfer Protocol]
0000 08 97 34 d0 48 00 4c 34 88 fa e0 dd 08 00 45 00
0010 00 98 cc 35 40 00 80 06 3f 5f ac 0f 34 65 67 66
0020 a6 f0 c6 d1 01 bb 7f 67 b3 78 b2 cc 9f d3 50 18
0030 30 c6 a6 47 00 00 17 03 03 00 6b 3e 1e ba 17 a4
0040 20 39 8d 15 0d f9 cd ba 2b bb 88 7d 7f 39 ab 32
0050 81 8f 23 9f 01 05 26 67 f7 93 c0 86 b0 b3 bb 77
0060 ef 91 22 df 27 63 af a5 db 25 2d 45 67 8c 34 00
0070 8b bf 75 47 ee c9 8e 7d 31 2f d7 07 3d ad 8f 06
0080 5b cf 46 10 35 c3 de 5b d7 a5 c6 2c 80 1d 3c fc
0090 3b 36 a6 f2 86 25 73 18 3b 56 1f 10 43 97 1b 4a
00a0 41 ad d7 ba e7 a1
No.: 1071 · Time: 14.848080 · Source: 172.15.52.101 · Destination: 103.102.166.240 · Protocol: TLSv1.2 · Length: 166 · Info: Application Data
Show packet bytes
Layout:
Vertical (Stacked)
Close
Help




Q1: What differences do you notice between HTTP and HTTPS traffic?

Answer:

When I checked the HTTP packets in Wireshark (from Task 1), I could clearly see the form data, like the username, password, and age, written in plain text.

With HTTPS packets, it was completely different. The actual content was hidden and only showed up as “Application Data,” which looked like random, scrambled letters and numbers.

So basically:

- HTTP = you can read everything clearly
- HTTPS = the data is locked/encrypted and unreadable

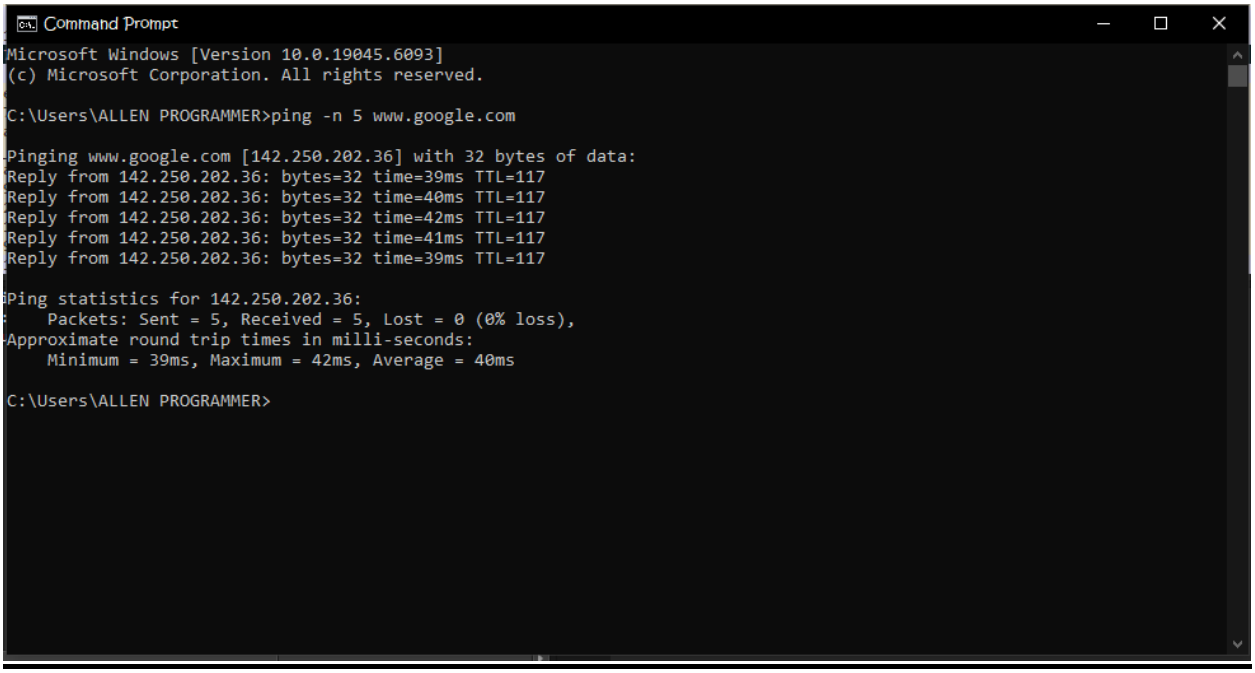
**Q2: Why is HTTPS considered more secure than HTTP?**

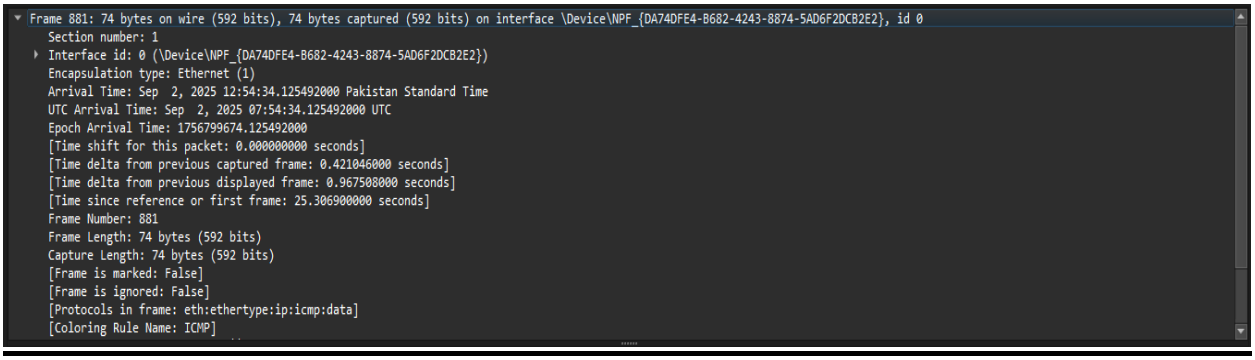
**Answer:**
HTTPS is safer because it uses encryption (SSL/TLS). This means that even if someone captures the packets, they won’t be able to read the actual data—it just looks like gibberish.

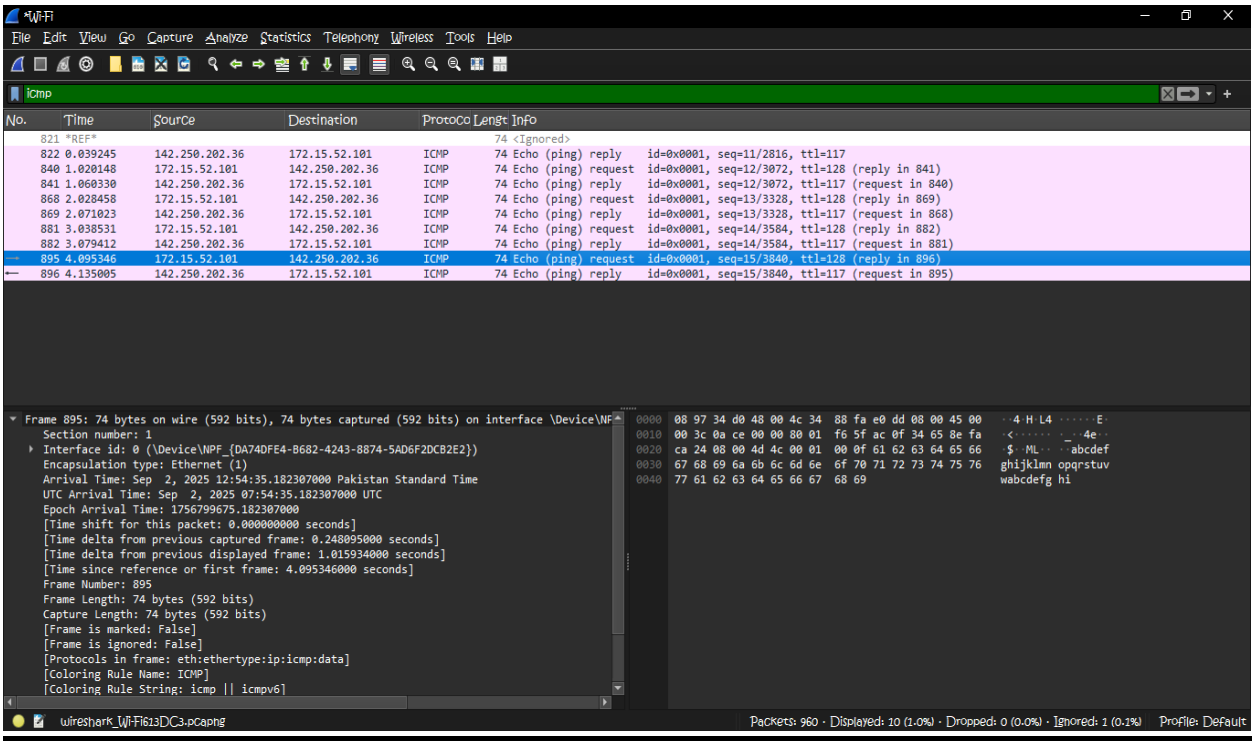
- It protects against **eavesdropping**, so hackers can’t see your personal details.
- It prevents **tampering**, so nobody can secretly change the data while it’s being sent.
- It also provides **authentication**, meaning you can be sure you’re talking to the real website and not a fake one.

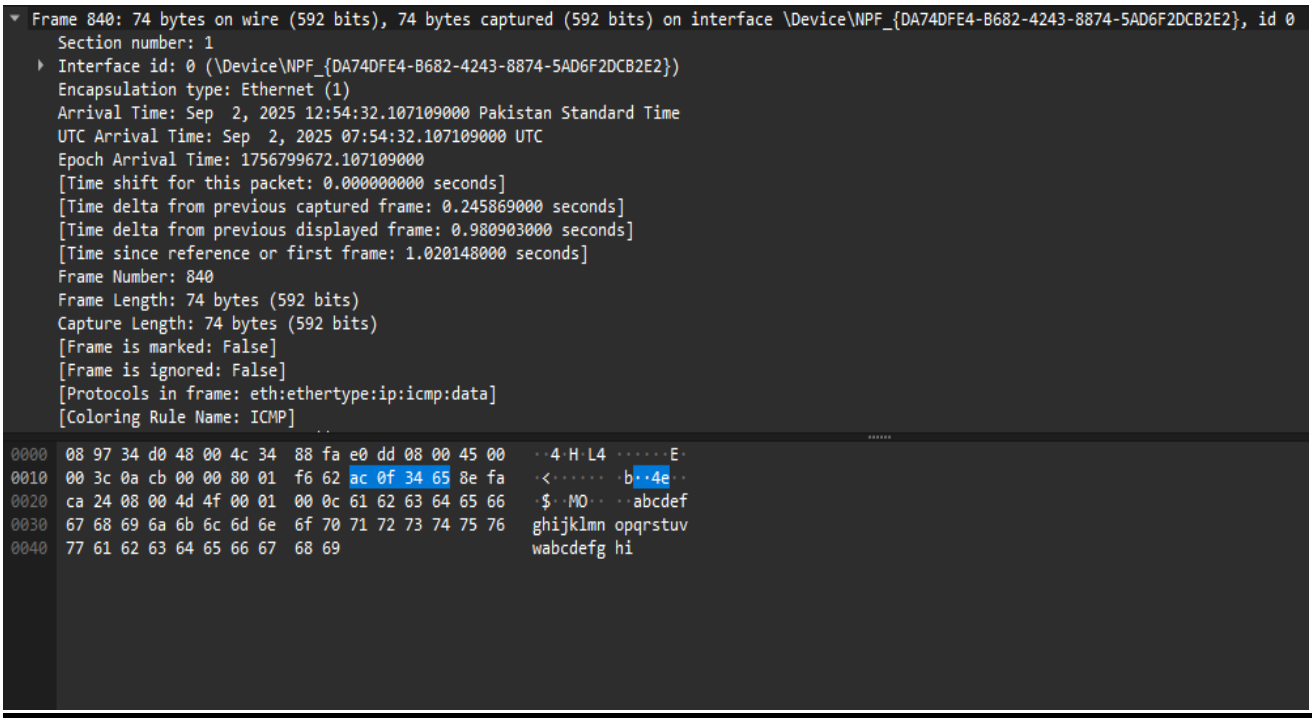
  That’s why HTTPS is used for things like online banking, shopping, and logging into accounts—it keeps the information private and secure.

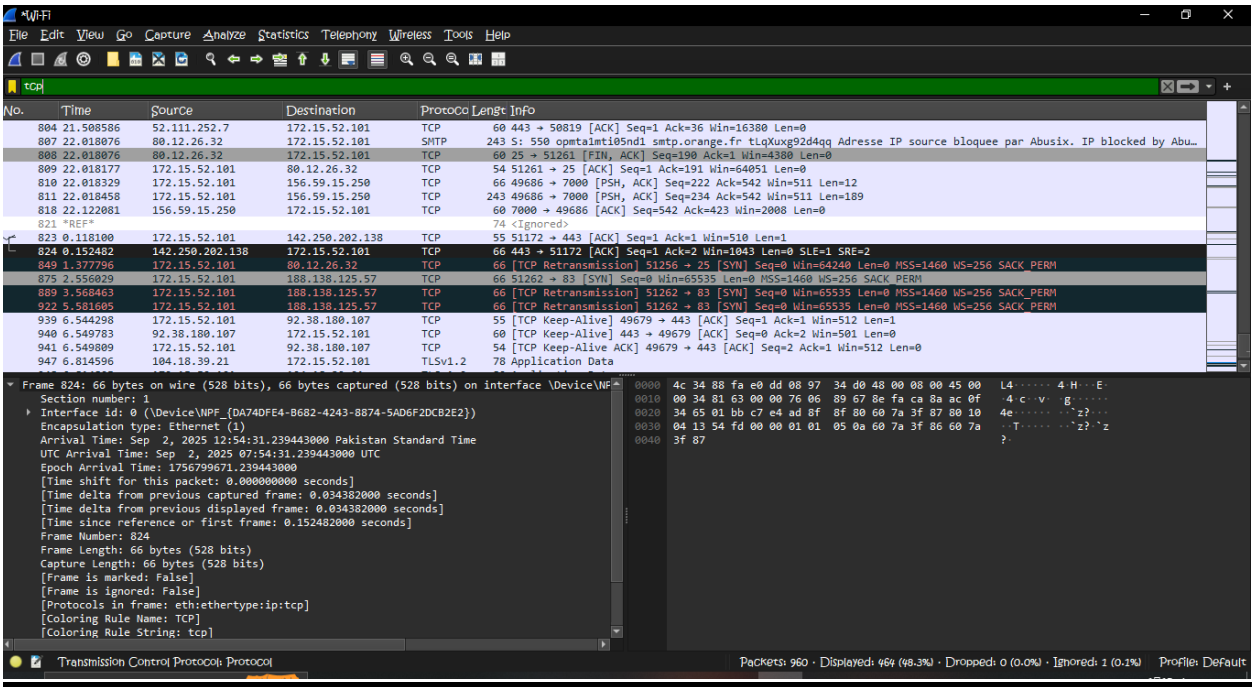
## Task# 03:


Command Prompt
Microsoft Windows [Version 10.0.19045.6093]
(c) Microsoft Corporation. All rights reserved.
C:\Users\ALLEN PROGRAMMER>ping -n 5 www.google.com
Pinging www.google.com [142.250.202.36] with 32 bytes of data:
Reply from 142.250.202.36: bytes=32 time=39ms TTL=117
Reply from 142.250.202.36: bytes=32 time=40ms TTL=117
Reply from 142.250.202.36: bytes=32 time=42ms TTL=117
Reply from 142.250.202.36: bytes=32 time=41ms TTL=117
Reply from 142.250.202.36: bytes=32 time=39ms TTL=117
Ping statistics for 142.250.202.36:
Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 39ms, Maximum = 42ms, Average = 40ms
C:\Users\ALLEN PROGRAMMER>


Frame 881: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{DA74DFE4-B682-4243-8874-5AD6F2DCB2E2}, id 0
Section number: 1
Interface id: 0 (\Device\NPF_{DA74DFE4-B682-4243-8874-5AD6F2DCB2E2})
Encapsulation type: Ethernet (1)
Arrival Time: Sep 2, 2025 12:54:34.125492000 Pakistan Standard Time
UTC Arrival Time: Sep 2, 2025 07:54:34.125492000 UTC
Epoch Arrival Time: 1756799674.125492000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 0.421046000 seconds]
[Time delta from previous displayed frame: 0.967508000 seconds]
[Time since reference or first frame: 25.306900000 seconds]
Frame Number: 881
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:icmp:data]
[Coloring Rule Name: ICMP]


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
No. Time Source Destination Protoco Lengt Info
821 *REF* 74 <Ignored>
822 0.039245 142.250.202.36 172.15.52.101 ICMP 74 Echo (ping) reply id=0x0001, seq=11/2816, ttl=117
840 1.020148 172.15.52.101 142.250.202.36 ICMP 74 Echo (ping) request id=0x0001, seq=12/3072, ttl=128 (reply in 841)
841 1.060330 142.250.202.36 172.15.52.101 ICMP 74 Echo (ping) reply id=0x0001, seq=12/3072, ttl=117 (request in 840)
868 2.028458 172.15.52.101 142.250.202.36 ICMP 74 Echo (ping) request id=0x0001, seq=13/3328, ttl=128 (reply in 869)
869 2.071023 142.250.202.36 172.15.52.101 ICMP 74 Echo (ping) reply id=0x0001, seq=13/3328, ttl=117 (request in 868)
881 3.038531 172.15.52.101 142.250.202.36 ICMP 74 Echo (ping) request id=0x0001, seq=14/3584, ttl=128 (reply in 882)
882 3.079412 142.250.202.36 172.15.52.101 ICMP 74 Echo (ping) reply id=0x0001, seq=14/3584, ttl=117 (request in 881)
895 4.095346 172.15.52.101 142.250.202.36 ICMP 74 Echo (ping) request id=0x0001, seq=15/3840, ttl=128 (reply in 896)
896 4.135005 142.250.202.36 172.15.52.101 ICMP 74 Echo (ping) reply id=0x0001, seq=15/3840, ttl=117 (request in 895)
Frame 895: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NF
Section number: 1
Interface id: 0 (\Device\NPF_{DA74DFE4-B682-4243-8874-5AD6F2DCB2E2})
Encapsulation type: Ethernet (1)
Arrival Time: Sep 2, 2025 12:54:35.182307000 Pakistan Standard Time
UTC Arrival Time: Sep 2, 2025 07:54:35.182307000 UTC
Epoch Arrival Time: 1756799675.182307000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 0.248095000 seconds]
[Time delta from previous displayed frame: 1.015934000 seconds]
[Time since reference or first frame: 4.095346000 seconds]
Frame Number: 895
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:icmp:data]
[Coloring Rule Name: ICMP]
[Coloring Rule String: icmp || icmpv6]
0000 08 97 34 d0 48 00 4c 34 88 fa e0 dd 08 00 45 00
0010 00 3c 0a ce 00 00 80 01 f6 5f ac 0f 34 65 8e fa
0020 ca 24 08 00 4d 4c 00 01 00 0f 61 62 63 64 65 66
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76
0040 77 61 62 63 64 65 66 67 68 69
wireshark_Wi-Fi613DC3.pcapng
Packets: 960 · Displayed: 10 (1.0%) · Dropped: 0 (0.0%) · Ignored: 1 (0.1%)
Profile: Default


Frame 840: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{DA74DFE4-B682-4243-8874-5AD6F2DCB2E2}, id 0
Section number: 1
Interface id: 0 (\Device\NPF_{DA74DFE4-B682-4243-8874-5AD6F2DCB2E2})
Encapsulation type: Ethernet (1)
Arrival Time: Sep 2, 2025 12:54:32.107109000 Pakistan Standard Time
UTC Arrival Time: Sep 2, 2025 07:54:32.107109000 UTC
Epoch Arrival Time: 1756799672.107109000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 0.245869000 seconds]
[Time delta from previous displayed frame: 0.980903000 seconds]
[Time since reference or first frame: 1.020148000 seconds]
Frame Number: 840
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:icmp:data]
[Coloring Rule Name: ICMP]
0000 08 97 34 d0 48 00 4c 34 88 fa e0 dd 08 00 45 00
0010 00 3c 0a cb 00 00 80 01 f6 62 ac 0f 34 65 8e fa
0020 ca 24 08 00 4d 4f 00 01 00 0c 61 62 63 64 65 66
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76
0040 77 61 62 63 64 65 66 67 68 69
ghijklmn opqrstuv
wabcdefg hi




**Arrival Time:** Sep 2, 2025 12:54:34.125492000 Pakistan Standard Time

**Frame Length:** 74 bytes (592 bits)

**First Echo Request:** Arrival Time: Sep 2, 2025 12:54:32.107109000 Pakistan Standard Time

**Last Echo Request:** Arrival Time: Sep 2, 2025 12:54:35.182307000 Pakistan Standard Time

**Difference in seconds**

From **32.107109** → **35.182307**

35.182307−32.107109=3.075198 seconds

**Time difference = 3.075198 seconds**

**Final Answer:**

The time difference is **4.135005 seconds** (≈ 4.14 seconds).

**Q1. How frequently are packets being sent?**

On Windows, the ping command usually sends one request every second. In my capture, the gap between each ICMP Echo Request was around 1 second, which matches this behavior.

**Q2. What is the average packet size, and how does it compare to other traffic types?**

The ping packets I captured were about 70–80 bytes each. That's pretty small compared to normal internet traffic. For example, web browsing over HTTPS often uses packets that are close to 1,500 bytes (the typical maximum size), while simple acknowledgment packets are smaller, usually around 50–60 bytes. So, ping packets are tiny compared to most real-world traffic.

**Q3. What transmission mode is used?**

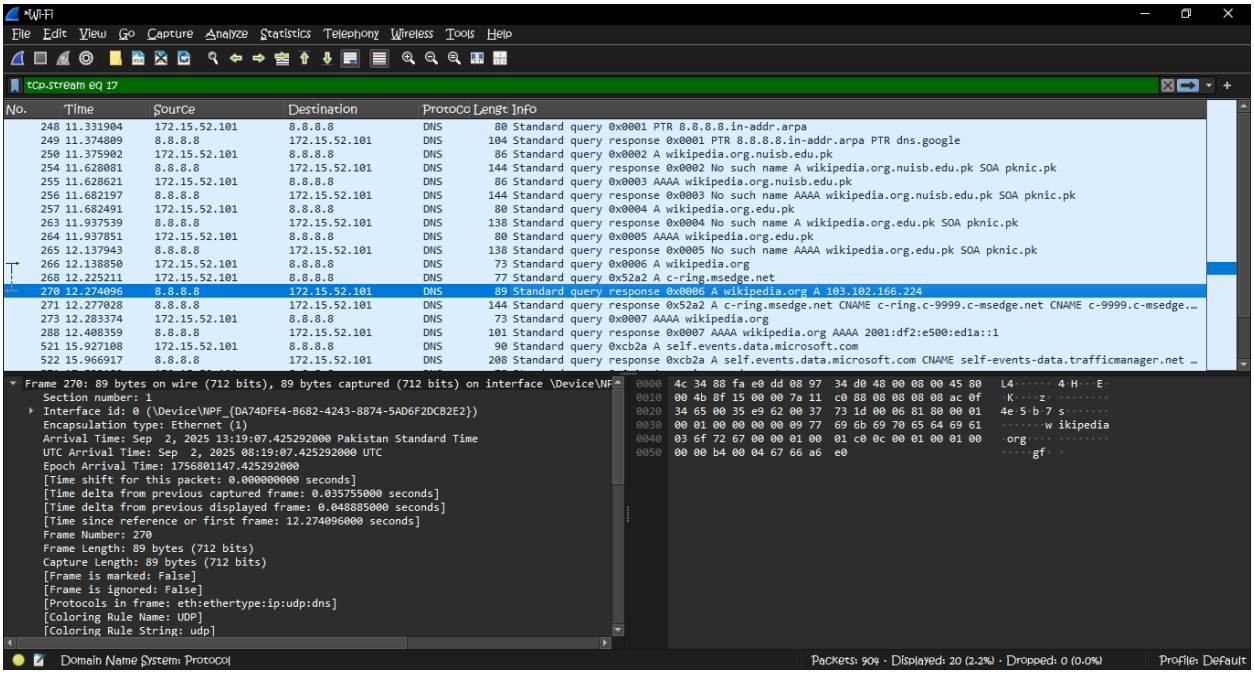
Data on networks is sent one bit at a time (serial transmission). On modern Ethernet, this is usually full-duplex, meaning data can flow in both directions at once. On Wi-Fi, it's typically half-duplex, so only one side talks at a time. Either way, the packets you see are transmitted serially.

## Task# 04:



```
        .... .0.. .... .... = Authoritative: Server is not an authority for domain
        .... ..0. .... .... = Truncated: Message is not truncated
        .... ...1 .... .... = Recursion desired: Do query recursively
        .... .... 1... .... = Recursion available: Server can do recursive queries
        .... .... .0.. .... = Z: reserved (0)
        .... .... ..0. .... = Answer authenticated: Answer/authority portion was not authenticated by the server
        .... .... ...0 .... = Non-authenticated data: Unacceptable
        .... .... .... 0000 = Reply code: No error (0)
    Questions: 1
    Answer RRs: 1
    Authority RRs: 0
    Additional RRs: 0
  ▼ Queries
    ▸ wikipedia.org: type A, class IN
  ▼ Answers
    ▸ wikipedia.org: type A, class IN, addr 103.102.166.224
    [Request In: 266]
    [Time: 0.135246000 seconds]

0000  4c 34 88 fa e0 dd 08 97  34 d0 48 00 08 00 45 80   L4······ 4·H···E·
0010  00 4b 8f 15 00 00 7a 11  c0 88 08 08 08 08 ac 0f   ·K····z· ········
0020  34 65 00 35 e9 62 00 37  73 1d 00 06 81 80 00 01   4e·5·b·7 s·······
0030  00 01 00 00 00 00 09 77  69 6b 69 70 65 64 69 61   ·······w ikipedia
0040  03 6f 72 67 00 00 01 00  01 c0 0c 00 01 00 01 00   ·org···· ········
0050  00 00 b4 00 04 67 66 a6  e0                       ·····gf· ·
```



Q1: Why is DNS critical for HTTP/HTTPS communication, and how does it relate to secure data transmission?

Answer:

DNS (Domain Name System) is like the phonebook of the internet. We type easy-to-remember names like *www.wikipedia.org*, but computers only understand numbers called IP addresses. DNS translates the name into the right IP so your browser knows where to connect.

For HTTPS, DNS is the very first step. Before a secure connection can even start, your computer has to find the correct server's IP through DNS. Once that's done, the HTTPS handshake kicks in — this is where the server proves its identity with a certificate, and all communication is encrypted and protected from tampering.